Salman Naveed

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Education

Master of Science in Electrical Engineering, *University of Stuttgart* 04/2023 – 12/2026 | Stuttgart, Germany
Theory coursework: Embedded Systems, Real-Time Software Engineering, Industrial Automation, Sensor Interface Circuits, Optical Fiber Communication, Deep Learning, Pattern Recognition, Memory Device Technology, Automation Systems Modelling & Analysis, Risk Assessment for Robotic Systems
Lab course: Microwave Engineering

Bachelor of Engineering in Electrical Engineering 09/2017 – 06/2021 | Karachi, Pakistan
National University of Sciences and Technology
Theory coursework: Embedded Systems, Digital Image Processing, Linear Control Systems, Communication Systems, RF Technology, Electrical Machines, Analog Electronics, Digital Electronics, Microprocessor Systems
Senior Design Project: Urban Air Quality Monitoring using Networked, Low-cost Remote Sensor Nodes

Experience

Werkstudent - Embedded Systems 06/2023 – Present | Stuttgart, Germany
Gesellschaft für Produktionsysteme (GPS) GmBH
- Led end-to-end hardware development lifecycle for 3 Industrial IoT product lines, driving projects from initial requirements and component selection through schematic design, multi-layer PCB layout, and validation
- Designed robust, multi-layer PCBs (up to 6 layers) in Altium Designer for industrial environments, successfully advancing through iterative phases to final production
- Designed compact 4-layer industrial IoT PCBs with isolated 24 V digital I/O, filtered 1–5 V analog sensor inputs and ESP32-S3 control, featuring TVS, reverse-polarity and short-circuit protection with EMC-aware layout
- Optimized hardware architectures for Design for Manufacturing (DFM) and assembly, including dedicated product line for optimized, cost-effective in-house low-volume production
- Supported firmware architecture and development in C/C++ on ESP32 and STM32 platforms for Industry 4.0 applications, managing concurrent tasks, real-time constraints, and edge processing
- Developed production-grade C/C++ embedded software enabling advanced edge use-cases, including digital twins, predictive maintenance, condition monitoring, and automated equipment testing
- Developed and implemented custom, safety-critical SLIP-based UART protocol in C/C++ for the ESP32 platform to handle high-volume, real-time communication in a motorized rail-based lifting system
- Optimized protocol firmware under strict timing and memory constraints, implementing deterministic buffer management and error-checking to guarantee zero data loss between industrial controllers and edge gateways
- Designed and implemented custom Hardware Abstraction Layer (HAL) in C/C++ on STM32H7 to encapsulate low-level QSPI and FreeRTOS APIs, enabling secure, thread-safe NVS persistent storage of device config and calibration data
- Developing internal and user-level documentation, test plans, qualification criteria for each new product
- Operating in a fast-paced startup environment with direct ownership of technical decisions, scope, and delivery timeline

Product Manager - Digital Payments & Onboarding Technologies 10/2021 – 03/2023 | Karachi, Pakistan
UBL - United Bank Limited
- Managed production-critical digital payment and onboarding platforms across 1,400+ locations, coordinating engineering, infrastructure, operations, SQA, compliance, and customer-facing teams
- Led cross-functional delivery across 4 major projects, coordinating technical and operational teams and escalating blockers to leadership for rapid resolution
- Managed 30+ concurrent medium-to-high-priority change requests, analysing technical and business impact, prioritizing work, coordinating resolution, and tracking implementation
- Supported rollout of infrastructure handling 250,000+ daily transactions at ~$10M volume, coordinating risk management, validation, and operational readiness
- Used SQL and advanced Excel extensively for operational analysis, validation, and issue investigation, supporting data-driven decisions across technical and business teams
- Acted as primary interface across technical teams, operations, external organizations, and regulators, demonstrating customer focus, urgency, structured escalation, and end-to-end ownership

Technical Skills

Hardware & Platforms: STM32, ESP32, AVR, NXP, Raspberry Pi, Intel x86
Software: C, C++, Python, MATLAB
OS & RTOS: FreeRTOS, ROS, Mbed OS, Embedded Linux, Bare-metal
Communication Protocols: UART, SPI, I²C, CAN, USB, Ethernet, MQTT, HTTP/S, MAVLink, COBS, SLIP
PCB Design: Altium Designer, KiCad (Schematic Capture, PCB Layout)
Development & Debugging: Git, GitHub, GitLab, Make, CMake, PlatformIO, JTAG/SWD, GDB, VSCode, STM32Cube Ecosystem (IDE, MX, Programmer, Monitor)
RF & Simulation: Keysight ADS (RF Filter & Coupler Design), Ansys HFSS, Ansys Electronics Desktop (Antenna Design), LTspice, MATLAB/Simulink
Lab & Measurement: Oscilloscopes, Vector Network Analyzers, Logic Analyzers, Spectrum Analyzers, Function Generators, Soldering & Rework, Automated Test, SCPI Setups
FPGA & HDL: Verilog, Intel Quartus (Coursework)
Office & Collaboration: Jira, Confluence, SharePoint, PowerBI, Miro, Trello, Microsoft Office
Mechanical CAD: Autodesk Inventor, Autodesk Fusion 360, SolidWorks

Languages

English
Full Professional Proficiency (C1)

German
Intermediate (B1)

Urdu
Native Speaker

Projects

XR-4: Autonomous Mobile Robotics Test Platform , *Personal Project* 08/2025 – Present

- Designed and built a 4WD rover test platform for autonomous robotics R&D, with sole ownership of mechanical, structural, electrical, firmware and system integration
- Utilized Fusion 360 for CAD with aluminum frame, 3D-printed parts and independent suspension
- Prototyped a three-board electrical system (onboard computer, power distribution board, handheld control and telemetry unit) with two ESP32s linked via ESP-NOW at <1 ms command latency
- Developing custom set of PCBs in KiCAD sponsored by PCBWay
- Developed C++ firmware (ESP-IDF, FreeRTOS) for MCPWM differential-drive control, telemetry and telecommands, and sensor integration (IMU, magnetometer, GPS, 16-bit ADC, power, temperature), with low-latency emergency stop
- Integrating an x86 ROS2 mission computer (AAEON UP²) with LIDAR and cameras for visual odometry, supervising the onboard computer via a custom COBS-based protocol

High Speed Polarization Tracking System , *Master Thesis* 06/2026 – 12/2026

- Design and development of a high-speed, closed-loop polarization tracking system delivering wavelength-independent linearly polarized light for characterizing few-mode and multicore fibers and couplers
- Designed a precision analog signal chain (DAC, amplification and buffering) and power conditioning to drive a high-voltage piezo driver for multi-axis polarization actuation
- Developed 2- and 4-layer mixed-signal PCBs integrating STM32 controller, analog and power stages, applying analog/digital partitioning and signal-integrity-aware routing for low-noise performance
- Performed make-or-buy analysis for polarization sensing and integrating a commercial polarimeter, avoiding multi-wavelength calibration challenges and unknown fiber birefringence inherent to an all-fiber in-house solution
- Implemented real-time STM32 firmware for closed-loop feedback control, targeting sub-millisecond response and complete 360° polarization rotation
- Developed a professional, cross-platform PC-based GUI client for instrument monitoring and control
- Validated and characterized the system using tunable fiber lasers, polarization controllers and lab instrumentation, supported by Python data-acquisition scripts and digital twins for performance analysis and troubleshooting

Group Delay Characterization of Planar Microwave Bandpass Filters, *Research project* 04/2025 – 11/2025

- Designed four planar microstrip band-pass filters (Butterworth and Chebyshev, orders 3–5) in coupled-line, hairpin and short-circuited stub topologies on Rogers RT/duroid 6002, at 2.4 GHz
- Executed the complete Keysight ADS workflow: low-pass prototype and J-inverter synthesis, LineCalc dimensioning, schematic simulation, Momentum EM simulation and optimization, and layout export for fabrication
- Connectorized the fabricated boards and characterized S11, S21 and group delay with Keysight FieldFox VNA using SOLT calibration, a 50 Ω thru-line reference to separate fixture losses, and MATLAB post-processing
- Achieved close agreement between measurement and EM simulation and identified over-etching as the root cause of the Butterworth filter's deviation from its design response.